



THE CRUCIAL PART FOR THE INDIAN ECOSYSTEM IS TO GET THE TATA-PSMC FAB AND THE FOUR OSAT PLANTS UP AND RUNNING WITH HIGH EFFICIENCY. THAT WILL MAKE INDIA'S PLANS CREDIBLE. THIS IS A 20-YEAR GAME, AND I WOULD NOT JUMP TO CONCLUSIONS BASED ON A FEW INDIVIDUAL BUSINESS DECISIONS.



— **PRANAY KOTASTHANE, DEPUTY DIRECTOR OF THE TAKSHASHILA INSTITUTION**

focus on Semiconductor Product Design, Semiconductor Manufacturing and Talent Development.”

Manufacturing alone will not ensure strategic autonomy in semiconductors. As India pivots to the next phase, design and innovation must also be prioritised alongside manufacturing. Dr Gupta advocates for an expanded DLI 2.0 scheme with an outlay of \$1 billion, representing approximately 10% of the 2022 incentives allocated to fabrication. “This will accelerate the development of important Indian chips by Indian companies and make India *Atma-Nirbhar* in this critical area to create strategic autonomy,” he states.

Dr Gupta also commended the fact that India is executing the world’s largest chip design education programme under the C2S initiative. “India has done very well in terms of providing the all-important chip design tools (EDA tools) from the leading global companies under the C2S programme to more than 280 institutions,” he said. Through this, India is enabling more than 50,000 students and 1000 faculty members across 280 institutes to gain training in all aspects of chip design.

Global collaborations are accelerating workforce development and technology transfer.

IMEC (Belgium) is partnering with Indian institutions on advanced manufacturing training, while companies such as Lam Research have launched local programmes to upskill engineers in chip fabrication technologies.

However, execution remains critical to bridging the gap between industry and academia.

Challenges persist in infrastructure development and in scaling domestic capabilities to meet global standards. These point to unique hurdles faced by individual models, as well as broader geopolitical influences.

Tariff, tensions, and fab

Investor caution towards semiconductor fabrication in India is closely linked to geopolitical instability. Foremost among these is the escalating tariff and technology war between the US and China, which has disrupted supply chains, increased the cost of critical inputs, and introduced long-term uncertainty in the chipmaking sector.

“Over the last few years, several fabrication models have been initiated globally. Some are not market demand-driven but are motivated by national security. Once all these fabs come online, they could face a serious underutilisation challenge. Thus, investors are wary

of committing to new fabrication, especially at a time when the US and China are engaged in a tariff war that may lead to inflation and lower demand. These models do not reflect India’s competitiveness, but rather the global situation,” Kotasthane notes.

In October 2022, the US imposed export controls on advanced chips and equipment to China. In retaliation, China restricted exports of gallium and germanium—key elements in semiconductors—banning them entirely to the US by December 2024. China currently controls 90% of global gallium and 60% of germanium production.

The war has escalated. As of April 2025, the US imposed sweeping tariffs, including a 145% levy on Chinese imports, citing national security concerns. China responded with a 125% tariff on American goods.

This has made investors increasingly cautious about committing to long-term semiconductor models worldwide, including in India. This hesitation is particularly evident in the reluctance to fund new fabrication facilities due to the persistent risk of additional trade restrictions or supply disruptions. Could this explain why Zoho could not find a suitable partner and Adani had to shelve its plans?

“For semiconductor fabs, individual models not taking off is not unusual. This segment has a long gestation period; companies and investors require a robust business case before committing,” says Kotasthane.

The government remains optimistic about the semiconductor pipeline, particularly in Gujarat and Assam. **EFY**

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